

Numbers

```
In [1]: 3
```

```
Out[1]: 3
```

```
In [2]: 5
```

```
Out[2]: 5
```

Arithmetic Operators

Operation	Operator	Description
Addition	+	Adds two values
Subtraction	-	Subtracts the right operand from the left
Multiplication	*	Multiplies two values
Division	/	Divides operands and returns a float (e.g., $5 / 2 = 2.5$)
Floor Division	//	Divides operands and returns an integer, rounding down (e.g., $5 // 2 = 2$)
Modulus	%	Returns the remainder of a division (e.g., $5 \% 2 = 1$)
Exponentiation	**	Raises one number to the power of another

```
In [18]: 5 + 3      # Addition
```

```
Out[18]: 8
```

```
In [19]: 8 - 3      # Subtraction
```

```
Out[19]: 5
```

```
In [20]: 5 * 4      # Multiplication
```

```
Out[20]: 20
```

```
In [21]: 8 / 2      # Division
```

```
Out[21]: 4.0
```

```
In [22]: 9 // 2     # Floor Division
```

```
Out[22]: 4
```

```
In [23]: 6 % 2      # Modulus
```

```
Out[23]: 0
```

```
In [24]: 3 ** 2     # Exponentiation
```

```
Out[24]: 9
```

Role of Underscore(_)

Python automatically stores the value of the last expression in the interpreter to a particular variable called "_".

We can also assign these value to another variable if you want.

```
In [14]: 5 + 4
```

```
Out[14]: 9
```

```
In [15]: _          # stores the result of the above expression
```

```
Out[15]: 9
```

```
In [16]: _ + 6
```

```
Out[16]: 15
```

```
In [26]: a = _      # assigning the value of _ to another variable
```

```
In [8]: a * 3
```

```
Out[8]: 21
```

BODMAS (or PEMDAS)

BODMAS Rule	Python Operator	Description
B rackets	()	Highest priority; forces order
O rders	**	Exponents/Powers
D ivision	/, //, %	Standard, floor division, and remainder
M ultiplication	*	Multiplication (equal priority with Division)
A ddition	+	Addition
S ubtraction	-	Subtraction

```
In [27]: 4 + 5 - 3 * 10      # (mas - mul, add, sub)
```

```
Out[27]: -21
```

```
In [28]: 4 + ( 5 - 10 ) * 3 # BODMAS
```

```
Out[28]: -11
```

```
In [29]: 4 / ( 5 - 10 ) * 3 # BODMAS
```

```
Out[29]: -2.4000000000000004
```

Example Calculation

Expression: $10 + 5 * (2 ** 3) - 6 / 2$

Step 1 (Brackets): $(2 ** 3) \rightarrow 8$

Step 2 (Multiplication): $5 * 8 \rightarrow 40$

Step 3 (Division): $6 / 2 \rightarrow 3.0$

Step 4 (Addition/Subtraction): $10 + 40 - 3.0 \rightarrow 50 - 3.0 \rightarrow 47.0$

Final Answer: 47.0

```
In [30]: 10 + 5 * (2 ** 3) - 6 / 2
```

```
Out[30]: 47.0
```

```
In [ ]:
```

```
In [ ]:
```